

# Open Issue for Discussion: Day-Ahead vs. Real-Time Coupling Band and Equilibrium Degeneracy at the Interconnection Ceiling

Parth Brahmhatt — University of Wisconsin–Madison

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## Abstract

In the iteration-free multi-agent receding-horizon market framework, the day-ahead (DAM) and real-time (RTM) stages currently use *different* numerical values for the point-of-common-coupling (PCC) import band  $[L_{\min}, L_{\max}]$ : the DAM uses  $[350, 600]$  kW while the RTM uses  $[100, 900]$  kW. Physically these are the *same* interconnection, so unifying them to  $[100, 900]$  seems obviously correct. We tried it. It **breaks the real-time solver**: production collapses and the explicit-map clearing fails on the vast majority of steps. This memo documents the experiment, the hard evidence, and the root cause — *generalized-Nash degeneracy when the aggregate import is pinned at the interconnection ceiling* — and lays out the design question we need to decide: whether the day-ahead ceiling should deliberately be a *conservative sub-band* of the physical envelope, or whether the real-time equilibrium should be reformulated to remain well-posed on the coupling boundary.

## 1 Background: the two coupling bands

Each 15-minute real-time step, the coalition’s grid imports obey a shared per-step band at the PCC,

$$L_{\min} \leq \sum_{i=1}^N p_{i,k} \leq L_{\max}, \quad k = 0, \dots, H-1, \quad (1)$$

where  $L_{\min}$  is the ISO minimum offer and  $L_{\max} = 0.75 \times 1200 = 900$  kW is the physical interconnection limit (75% of the 1200 kW fleet nameplate). The same band, at hourly resolution, constrains the day-ahead schedule that produces the anchor  $p_{i,h}^{\text{DA}}$ .

**The inconsistency.** In code, the RTM enforces  $[L_{\min}, L_{\max}] = [100, 900]$  (`rhg.mppq.py`) but the DAM enforced  $[350, 600]$  (inherited from an earlier single-step demo). Because  $L_{\min}, L_{\max}$  are physical/market constants, the two stages “should” use the same numbers. The natural fix is to set the DAM to  $[100, 900]$  as well. This memo is about what happened when we did.

## 2 The experiment: unify the DAM band to $[100, 900]$

We changed only the DAM band ( $350/600 \rightarrow 100/900$ ); the RTM was already 100/900. The day-ahead distributed ADMM still *converges* and produces a valid schedule — the DAM itself is fine. Table 1 shows the resulting day-ahead decision for one representative day.

Table 1: DAM grid-import decision  $p_{i,h}^{\text{DA}}$  (kW), ERCOT 2025-04-01, band [100, 900]. Only selected hours shown. The aggregate is *pinned at the 900 ceiling* on 8 of 24 hours, and the two grid agents sit at their rated power (250/200) most hours.

| hr | PEM_EI | ALK   | PEM_PV | PEM_PV2 | PEM_Wd | PEM_Wd2 | $\sum p$   | $\lambda_{\text{DA}}$ |
|----|--------|-------|--------|---------|--------|---------|------------|-----------------------|
| 1  | 240.9  | 182.1 | 125.0  | 125.0   | 113.5  | 113.5   | <b>900</b> | 2                     |
| 4  | 184.9  | 126.1 | 114.7  | 114.7   | 179.8  | 179.8   | <b>900</b> | -14                   |
| 7  | 205.9  | 147.1 | 125.0  | 125.0   | 148.5  | 148.5   | <b>900</b> | 20                    |
| 10 | 248.5  | 189.7 | 77.4   | 77.4    | 153.4  | 153.4   | <b>900</b> | 24                    |
| 12 | 202.0  | 143.1 | 89.8   | 89.8    | 187.7  | 187.7   | <b>900</b> | 10                    |
| 22 | 102.5  | 43.7  | 99.0   | 99.0    | 97.3   | 97.3    | 539        | 50                    |
| 23 | 219.8  | 161.0 | 119.4  | 119.4   | 140.2  | 140.2   | <b>900</b> | 24                    |

Two features of this schedule matter:

1. **The aggregate hugs the ceiling.**  $\sum_i p_{i,h}^{\text{DA}} \in [539, 900]$ , hitting  $L_{\max} = 900$  on a third of the hours. Under the old [350, 600] band the aggregate was [350, 600] — *always*  $\geq 300$  kW below the interconnection limit.
2. **Hydrogen is massively over-produced (156–182% of target).** The daily  $\text{H}_2$  target is a *floor* ( $\geq 0.55$  capacity factor), not a ceiling. With cheap ERCOT power (often  $\lambda < \$60/\text{MWh}$  break-even) and  $\text{H}_2$  valued at \$3/kg, buying at the full ceiling is *profitable*, so the economically-optimal DAM buys as much as the wire allows. The tighter 600 ceiling previously made that physically impossible.

### 3 What broke: the real-time solver collapses

The RTM tracks  $p^{\text{DA}}$  (soft  $\gamma$ -penalty) and re-optimizes on live prices. With the ceiling-hugging anchor, *every* RTM step is driven to  $\sum_i p_{i,k} \approx 900$  — the coupling constraint (1) binds. The consequences (Tables 2–3):

Table 2: RTM outcome, both validation weeks: DAM band [350, 600] (working) vs. [100, 900]. A “miss” is a step the explicit map cannot resolve.

| Metric                        | DAM [350, 600] (interior anchor) |          | DAM [100, 900] (ceiling anchor) |                     |
|-------------------------------|----------------------------------|----------|---------------------------------|---------------------|
|                               | April                            | July     | April                           | July                |
| Map misses / week (of 672)    | <b>0</b>                         | <b>0</b> | <b>322</b>                      | up to <b>96/day</b> |
| Weekly $\text{H}_2$ vs target | 125%                             | 119%     | <b>71%</b>                      | <b>0–121%</b>       |
| Map = centralized (max)       | $\leq 5.4 \times 10^{-4}$ kW     | —        | <b>91 kW</b>                    | <b>22 kW</b>        |

On several July days *every* step failed ( $\text{H}_2 = 0\%$ : the coalition produced nothing). To see the mechanism we logged each step (Table 3): we first try the previous critical-region (CR) combination, then walk the FACET neighbour graph out to 3 hops (1st/2nd/3rd neighbour), and only if that fails do we fall back to iterative ADMM.

Table 3: Per-step trace, 2025-04-01, DAM band [100, 900]. The map walk resolves the step at  $t = 0, 17$ ; *every other step* exhausts the 3-hop neighbour walk and falls back to ADMM — which itself runs the full 3000 iterations *without converging* ( $\sim 6$  s/step).  $\sum p$  stays at the ceiling throughout.

| step $t$ | resolution                                       | wall time       | $\sum_i p_{i,0}$ [kW]     |
|----------|--------------------------------------------------|-----------------|---------------------------|
| 0        | map walk (1 combo checked)                       | 0.9 ms          | 872                       |
| 1        | <b>ADMM fallback (3000 iters, not converged)</b> | 6319 ms         | 868                       |
| 2        | ADMM fallback (3000 iters)                       | 6148 ms         | 871                       |
| 4        | ADMM fallback (3000 iters)                       | 6156 ms         | <b>900</b>                |
| $\vdots$ | $\vdots$                                         | $\vdots$        | $\vdots$                  |
| 17       | map walk (1 combo checked)                       | 2.4 ms          | 894                       |
| 18       | ADMM fallback (2982 iters)                       | 5942 ms         | 880                       |
| $\vdots$ | (94 of 96 steps fall back)                       | $\sim 6$ s each | $\approx 855\text{--}900$ |

So **neither** solver works at the ceiling: the explicit map finds no self-consistent CR combination within 3 hops, *and* the ADMM fallback fails to converge in 3000 iterations. This is not an implementation slip in the neighbour walk — the walk correctly reports “no valid combo” and escalates. The equilibrium itself is the problem.

## 4 Root cause: GNE degeneracy on the coupling boundary

### Why the equilibrium is ill-posed when $\sum_i p_{i,k} = L_{\max}$

Interior to the band, the shared constraint (1) is *inactive*: each agent’s best response depends on the others only through the aggregate it sees, the game is an exact potential game, and its *variational* GNE is a single-valued, piecewise-affine map of the parameters — exactly what the explicit multiparametric map represents, and why it matched the centralized optimum to  $\leq 10^{-4}$  kW under the old band.

When the aggregate is pinned at  $L_{\max}$ , the coupling *binds*. At a binding shared constraint the variational GNE requires a *common* Lagrange multiplier enforced across all  $N$  agents simultaneously (the coupled KKT stationarity). The per-agent explicit maps each carry their *own* view of the coupling, so their fixed point on the boundary is a *degenerate vertex* where (i) the block-linear equilibrium system can become singular / rank-deficient, and (ii) multiple non-variational “corner” equilibria coexist with the variational one. The 1-hop min-potential refinement that normally selects the variational branch has no self-consistent combo to refine, so the walk returns nothing. Independently, ADMM’s convergence rate collapses at the degenerate active set (consistent with the pre-existing note that ADMM was “unreliable/unconverged at these settings”), so it too fails.

This is a *known* limitation, already recorded in the detailed report:

*“Degenerate vertices. Where a capacity box and the coupling activate together, the 1-hop variational refinement can leave a bounded residual;  $L_{\min} = 100$  keeps most operation interior, so it is small, and there were zero misses.”*

The phrase “keeps most operation interior” is the whole point. The tighter 600 DAM ceiling was **keeping real-time dispatch in the interior**, away from the boundary where the method

degenerates. Under the 900 DAM ceiling, “most operation” is no longer interior — it is *on* the boundary, every step — and the method has nothing to fall back on.

### The one-line finding

The DAM ceiling being lower than the physical interconnection limit is **not a leftover bug**. It is load-bearing: it holds the real-time equilibrium in the well-posed interior. Unifying the DAM ceiling to the full 900 kW is *incompatible* with the current explicit-map GNE method.

## 5 The question for discussion

### What we need to decide

Is it acceptable (and defensible in the paper) to let the *day-ahead commitment* use a conservative import ceiling  $L_{\max}^{\text{DA}} < L_{\max}^{\text{RT}} = 900$ , so that real-time dispatch stays interior and the iteration-free equilibrium remains well-posed? Or must the real-time equilibrium be reformulated to remain single-valued *on* the coupling boundary?

The options, with trade-offs:

**Option A — conservative day-ahead ceiling (recommended).** Keep  $L_{\min} = 100$  shared by both stages, but set the DAM ceiling to a conservative  $L_{\max}^{\text{DA}} \approx 600$  kW, documented as an intentional design choice: *the coalition commits day-ahead to a conservative import ceiling and reserves the real-time headroom up to the full 900 kW interconnection*. This is a legitimate, common two-settlement posture (day-ahead positions are routinely conservative relative to physical limits) and it *also* keeps the equilibrium interior and unique. Restores the validated results (0 misses, 119–125%  $\text{H}_2$ ,  $\leq 5.4 \times 10^{-4}$  kW). *Cost*: the paper must state and justify the two ceilings rather than claim a single physical band; the coalition forgoes some cheap-power arbitrage in the top 300 kW of the wire.

**Option B — make the RT equilibrium well-posed on the boundary.** Regularize the coupling (e.g. a small quadratic penalty / soft constraint, or an explicit common-multiplier formulation) so the variational GNE stays single-valued when  $\sum p = L_{\max}$ . This lets the DAM legitimately use the full 900 ceiling. *Cost*: it changes the RTM optimization model, requires re-deriving and re-solving the per-agent multiparametric maps offline, and re-validating accuracy; it may also enlarge the critical-region count.

**Option C — cap day-ahead over-production.** Add a day-ahead  $\text{H}_2$  *ceiling* (or an import cap below 900) so the DAM stops buying at the interconnection limit. *Cost*: this is economically equivalent to Option A (it is just another way of keeping the anchor off the ceiling) but is harder to justify — it forbids profitable production rather than making a conservative commitment.

**Option D — accept ADMM at binding-coupling steps.** Not viable as-is: the evidence shows ADMM *does not converge* at these degenerate steps (3000 iterations, non-converged), and it would forfeit the “iteration-free, one broadcast per step” contribution that is the paper’s core claim.

**Recommendation.** Option A: keep the conservative day-ahead ceiling, share  $L_{\min} = 100$ , and document the two ceilings as a deliberate conservative-commitment design. It is the only option that preserves the validated iteration-free results without a modeling overhaul, and it is honest about *why* the day-ahead position is conservative. Option B is the “right” long-term fix if a reviewer insists the day-ahead stage must be allowed to the full physical limit, but it is a research task (re-derive + re-validate the maps), not a constant change.

## Appendix: reproduction

DAM decision table: `dam.solve_dam_admm` on 2025-04-01 forecast (band set in `rhg_mqp.L_MIN/L_MAX`). Weekly RTM: `simple_game/rhg_week.py`. Per-step trace: `rhg_online.solve_step` with per-step data-transfer logging (map-walk step = 1 broadcast; fallback step = ADMM iteration count). All ERCOT 2025 data in `data/ercot/`.